UNIT – 2

CLASS ACTIVITY – 01 **20/07/2026**

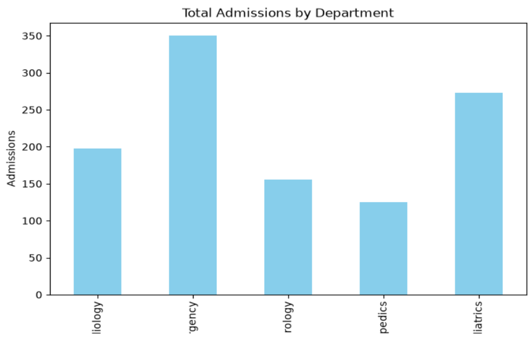
1. Hospital Patient Admission Analysis

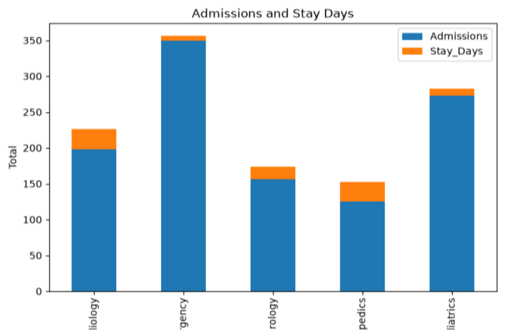
Python Code:

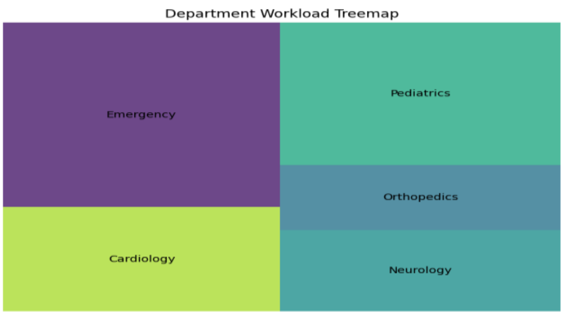
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
df = pd.read_excel("hospital.xlsx")
plt.figure(figsize=(8,5))
dept = df.groupby("Department")["Admissions"].sum()
dept.plot(kind='bar', color='skyblue')
plt.title("Total Admissions by Department")
plt.xlabel("Department")
plt.ylabel("Admissions")
plt.show()
stack = df.pivot_table(index="Department",
                       values=["Admissions","Stay_Days"],
                       aggfunc="sum")
stack.plot(kind="bar", stacked=True, figsize=(8,5))
plt.title("Admissions and Stay Days")
plt.ylabel("Total")
plt.show()
import squarify

plt.figure(figsize=(8,6))
sizes = dept.values
labels = dept.index

squarify.plot(
    sizes=sizes,
    label=labels,
    alpha=0.8
```

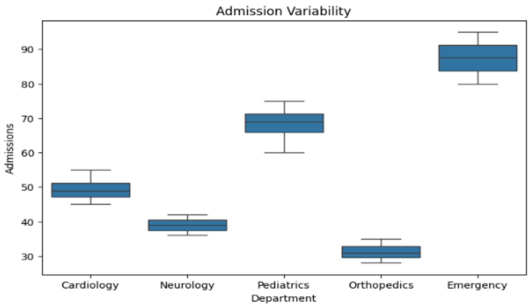
```
)
plt.title("Department Workload Treemap")
plt.axis("off")
plt.show()

plt.figure(figsize=(8,5))
sns.boxplot(x="Department", y="Admissions", data=df)
plt.title("Admission Variability")
plt.show()
plt.figure(figsize=(8,5))
sns.histplot(df["Admissions"], bins=8, kde=True, color="green")
plt.title("Admission Distribution")
plt.xlabel("Admissions")
plt.show()
```

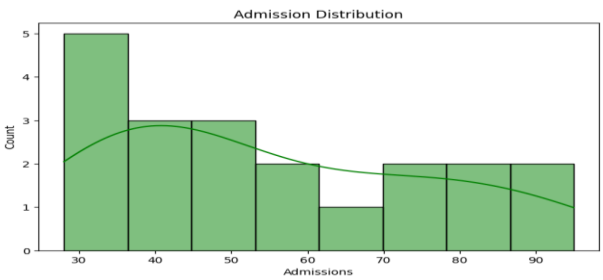
Output:







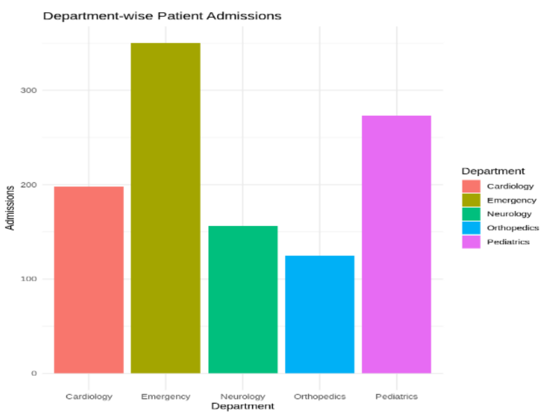


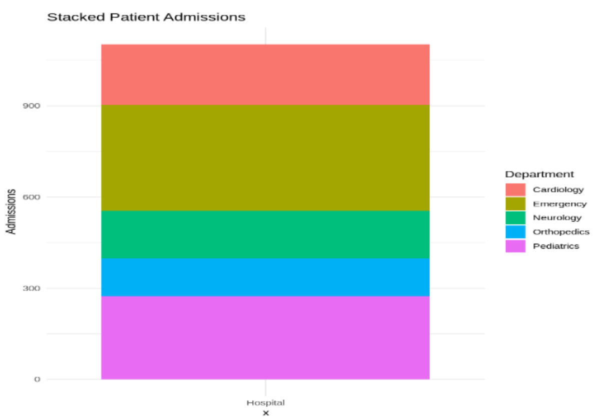


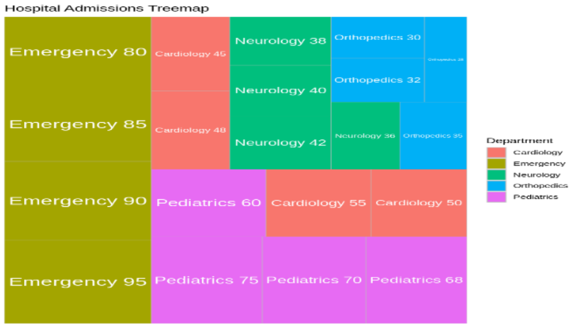
R code:

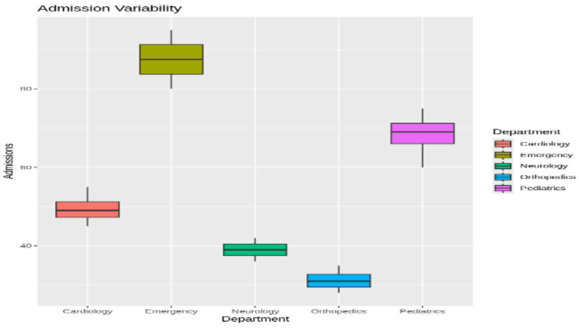
```
hospital <- read_excel("hospital.xlsx")
ggplot(hospital, aes(x = Department, y = Admissions, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Department-wise Patient Admissions") +
  theme_minimal()
ggplot(hospital, aes(x = "Hospital", y = Admissions, fill = Department)) +
  geom_bar(stat = "identity") +
  labs(title = "Stacked Patient Admissions") +
  theme_minimal()
ggplot(hospital, aes(area = Admissions, fill = Department,
                     label = paste(Department, Admissions))) +
  geom_treemap() +
  geom_treemap_text(colour="white", place="centre") +
  labs(title="Hospital Admissions Treemap")
ggplot(hospital, aes(x = Department, y = Admissions, fill = Department)) +
  geom_boxplot() +
  labs(title="Admission Variability")
ggplot(hospital, aes(Admissions)) +
  geom_histogram(binwidth = 20, fill = "steelblue") +
  labs(title="Admission Distribution")
```

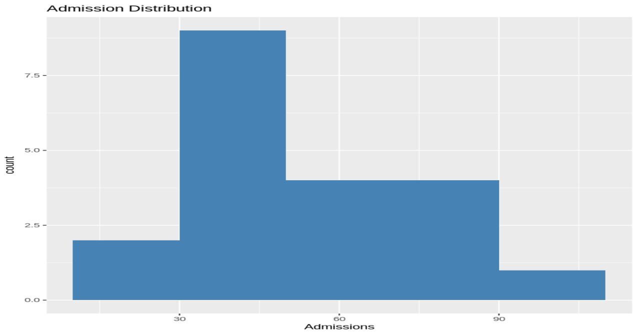
Output:











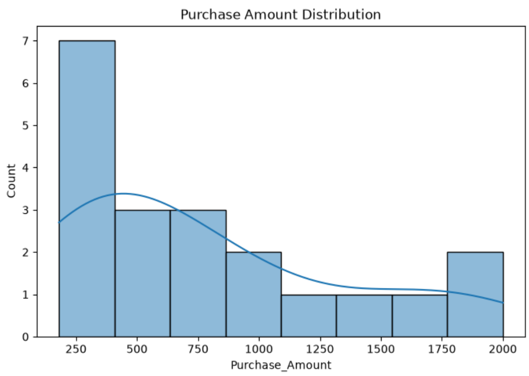
2. E-Commerce Customer Spending Behavior

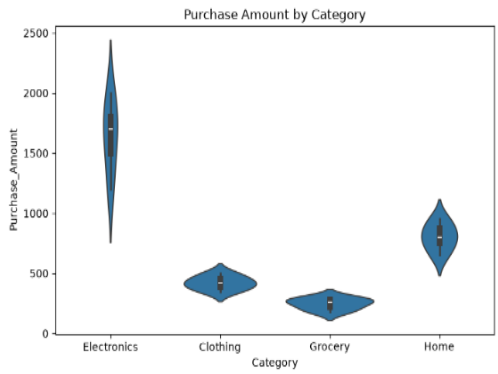
Python Code:

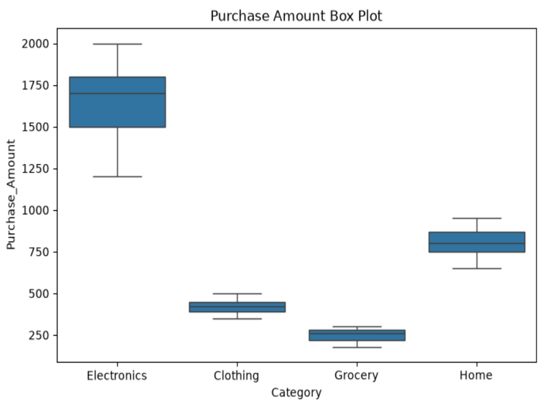
```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Read Dataset

df = pd.read_excel("ECommerce_Customer_Spending.xlsx")

# Histogram

plt.figure(figsize=(8,5))

sns.histplot(df["Purchase_Amount"], bins=8, kde=True)

plt.title("Purchase Amount Distribution")

plt.show()

# Violin Plot

plt.figure(figsize=(8,5))

sns.violinplot(x="Category", y="Purchase_Amount", data=df)

plt.title("Purchase Amount by Category")

plt.show()

# Box Plot

plt.figure(figsize=(8,5))

sns.boxplot(x="Category", y="Purchase_Amount", data=df)
```

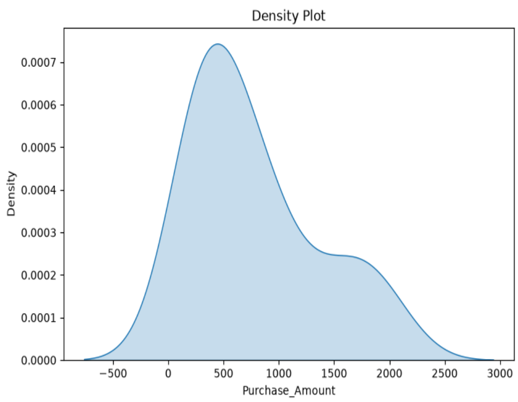
```
plt.title("Purchase Amount Box Plot")
plt.show()
# Density Plot
plt.figure(figsize=(8,5))
sns.kdeplot(df["Purchase_Amount"], fill=True)
plt.title("Density Plot")
plt.show()
# Bar Chart
avg = df.groupby("Category")["Purchase_Amount"].mean()
plt.figure(figsize=(8,5))
avg.plot(kind="bar")
plt.title("Average Spending by Category")
plt.ylabel("Average Purchase")
plt.show()
```

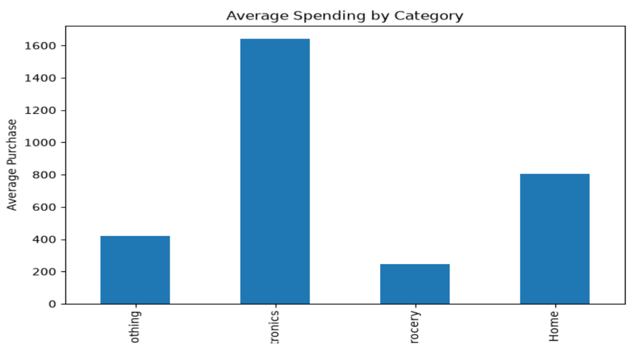
Output:









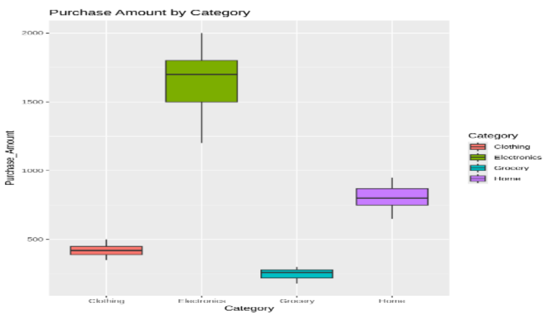


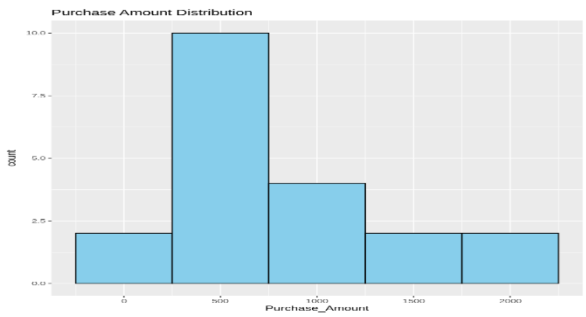
R code:

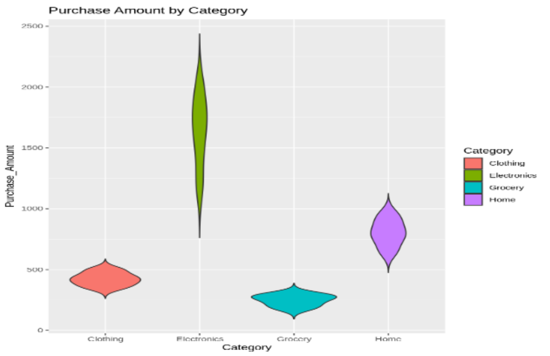
```
library(readxl)

library(ggplot2)

# Read Excel file

data <- read_excel("ECommerce_Customer_Spending.xlsx")

# 1. Histogram - Purchase Amount

ggplot(data, aes(x = Purchase_Amount)) +

  geom_histogram(binwidth = 500,

                 fill = "skyblue",

                 color = "black") +

  ggtitle("Purchase Amount Distribution")

# 2. Violin Plot

ggplot(data,

       aes(x = Category,

           y = Purchase_Amount,

           fill = Category)) +

  geom_violin(trim = FALSE) +

  ggtitle("Purchase Amount by Category")

# 3. Box Plot

ggplot(data,

       aes(x = Category,

           y = Purchase_Amount,

           fill = Category)) +

  geom_boxplot() +

  ggtitle("Purchase Amount by Category")
```

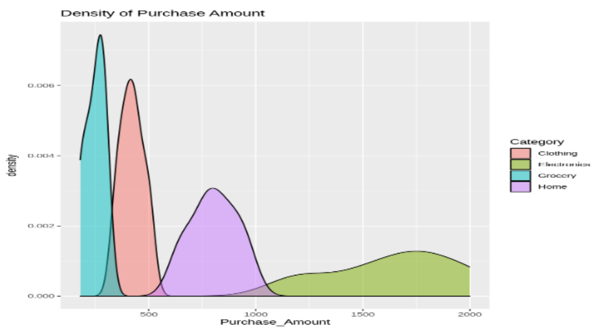
```
# 4. Density Plot
ggplot(data,
       aes(x = Purchase_Amount,
           fill = Category)) +
  geom_density(alpha = 0.5) +
  ggtitle("Density of Purchase Amount")
# 5. Bar Chart - Total Purchase Amount
ggplot(data,
       aes(x = Category,
           y = Purchase_Amount,
           fill = Category)) +
  stat_summary(fun = sum,
               geom = "bar") +
  ggtitle("Total Purchase Amount by Category")
```

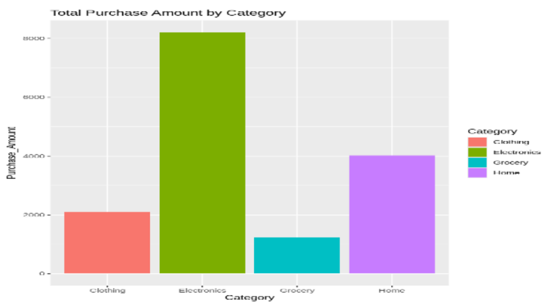
Output:











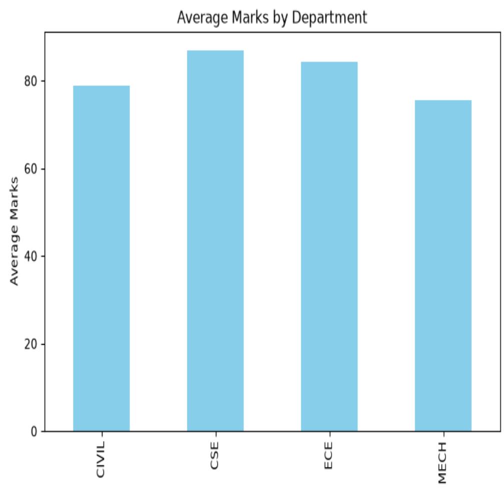
3. University Examination Performance Dashboard

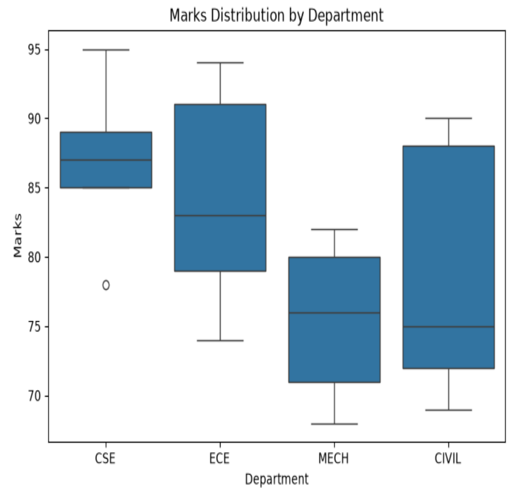
Python Code:

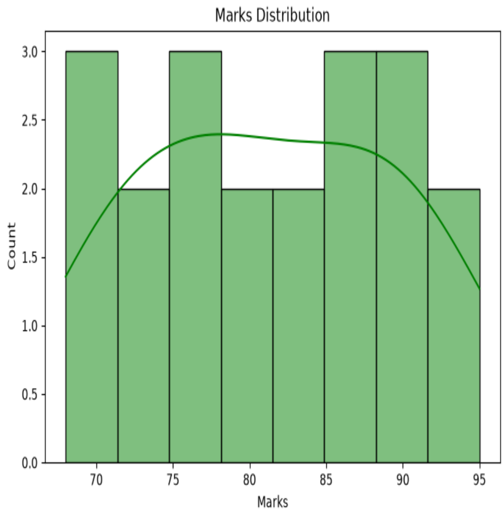
```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Read Dataset

df = pd.read_excel("University_Exam_Performance.xlsx")

# 1. Bar Chart - Average Marks

plt.figure(figsize=(8,5))

avg_marks = df.groupby("Department")["Marks"].mean()

avg_marks.plot(kind="bar", color="skyblue")

plt.title("Average Marks by Department")

plt.xlabel("Department")

plt.ylabel("Average Marks")

plt.show()

# 2. Box Plot

plt.figure(figsize=(8,5))

sns.boxplot(x="Department", y="Marks", data=df)

plt.title("Marks Distribution by Department")

plt.show()

# 3. Histogram

plt.figure(figsize=(8,5))

sns.histplot(df["Marks"], bins=8, kde=True, color="green")

plt.title("Marks Distribution")

plt.xlabel("Marks")

plt.show()

# 4. Violin Plot

plt.figure(figsize=(8,5))

sns.violinplot(x="Department", y="Marks", data=df)

plt.title("Marks Distribution using Violin Plot")

plt.show()

# 5. Scatter Plot
```

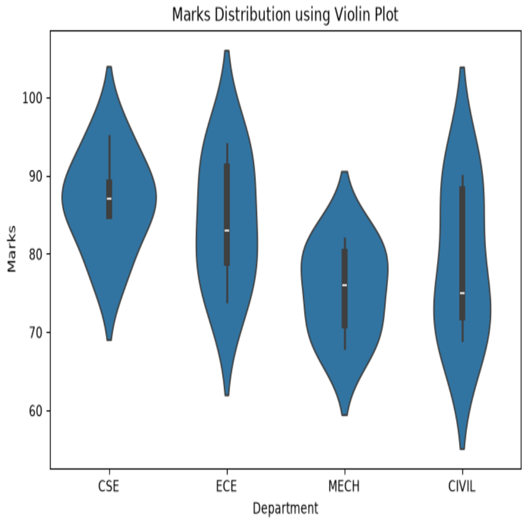
```
plt.figure(figsize=(8,5))
sns.scatterplot(
    x="Department",
    y="Marks",
    hue="Class",
    s=100,
    data=df
)
plt.title("Student Marks by Department")
plt.show()
```

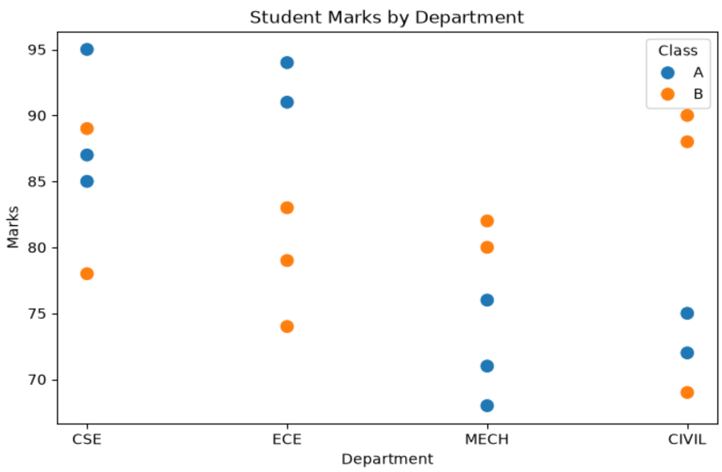
Output:


Average Marks by Department
Average Marks
0
20
40
60
80
CIVIL
CSE
ECE
MECH


Marks Distribution by Department
Marks
70
75
80
85
90
95
CSE
ECE
MECH
CIVIL
Department


Marks Distribution
Count
0.0
0.5
1.0
1.5
2.0
2.5
3.0
70
75
80
85
90
95
Marks

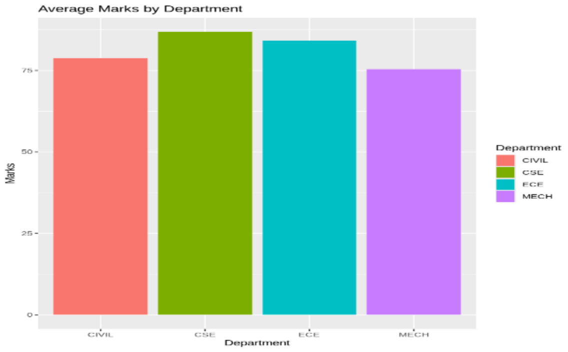

Marks Distribution using Violin Plot
Marks
60
70
80
90
100
CSE
ECE
MECH
CIVIL
Department

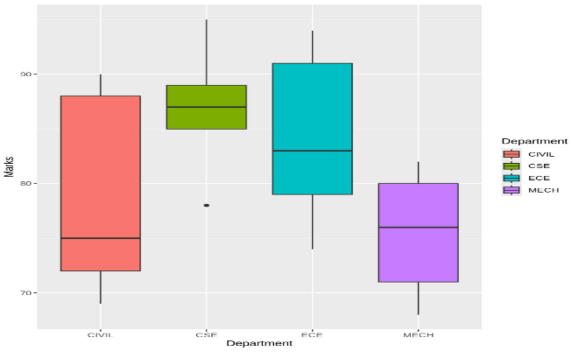

Student Marks by Department
Marks
70
75
80
85
90
95
CSE
ECE
MECH
CIVIL
Department
Class
A
B

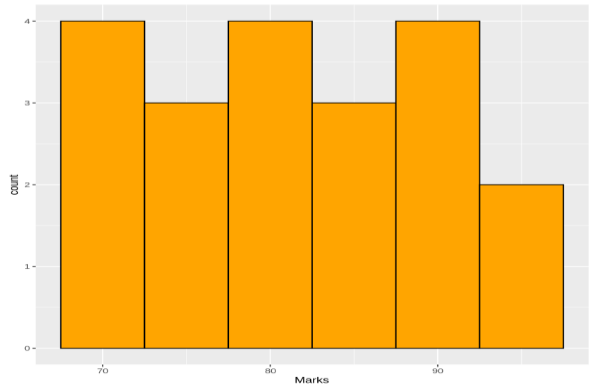
R code:

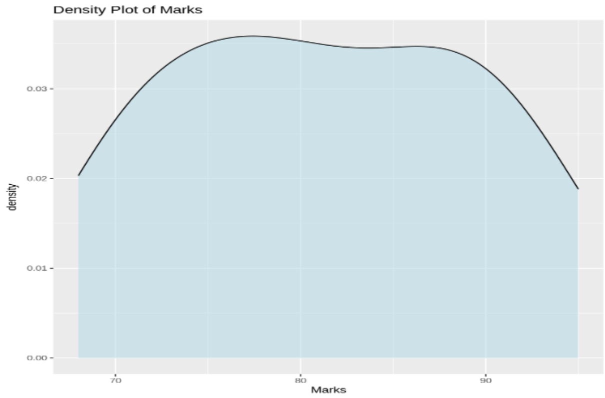
```
library(readxl)
library(ggplot2)
exam <- read_excel("University_Exam_Performance.xlsx")
# 1. Bar Chart - Average Marks by Department
ggplot(exam, aes(x = Department, y = Marks, fill = Department)) +
  stat_summary(fun = mean, geom = "bar") +
  ggtitle("Average Marks by Department")
# 2. Box Plot
ggplot(exam, aes(x = Department, y = Marks, fill = Department)) +
  geom_boxplot() +
  ggtitle("Marks Distribution by Department")
# 3. Histogram
ggplot(exam, aes(x = Marks)) +
  geom_histogram(binwidth = 5, fill = "orange", color = "black") +
  ggtitle("Marks Distribution")
# 4. Density Plot
ggplot(exam, aes(x = Marks)) +
  geom_density(fill = "lightblue", alpha = 0.5) +
  ggtitle("Density Plot of Marks")
# 5. Scatter Plot
ggplot(exam, aes(x = Student_ID, y = Marks, color = Department)) +
  geom_point(size = 4) +
  ggtitle("Student Marks by Department") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

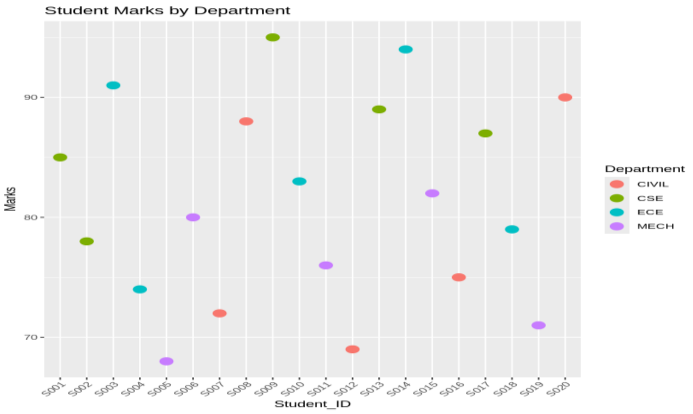
Output:











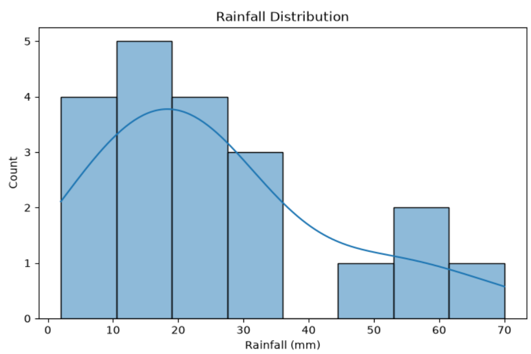
4. Climate and Rainfall Distribution Study

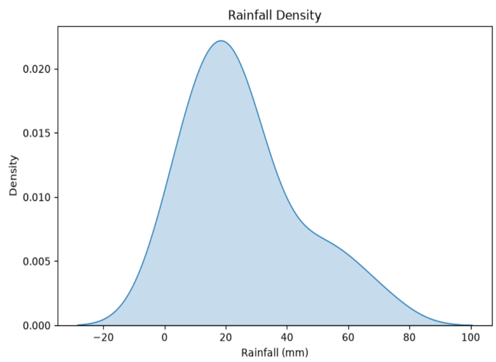
Python Code:

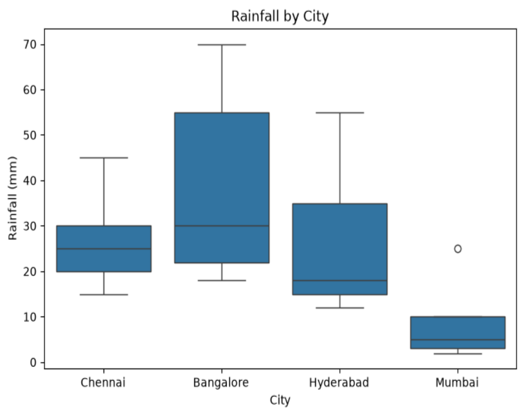
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
# Read Dataset
df = pd.read_excel("Climate_Rainfall_Data.xlsx")
# 1. Histogram
plt.figure(figsize=(8,5))
sns.histplot(df["Rainfall (mm)"], bins=8, kde=True)
plt.title("Rainfall Distribution")
plt.show()
# 2. Density Plot
plt.figure(figsize=(8,5))
sns.kdeplot(df["Rainfall (mm)"], fill=True)
plt.title("Rainfall Density")
plt.show()
# 3. Box Plot
```

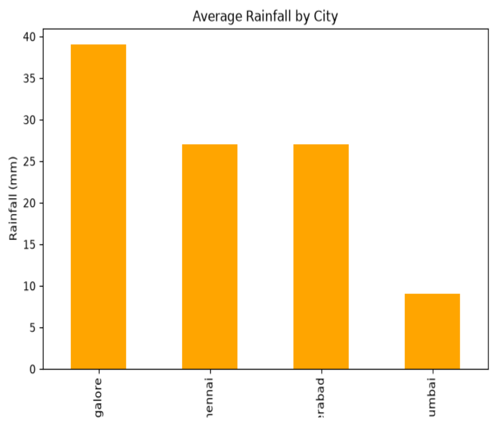
```
plt.figure(figsize=(8,5))
sns.boxplot(x="City", y="Rainfall (mm)", data=df)
plt.title("Rainfall by City")
plt.show()
# 4. Bar Chart
plt.figure(figsize=(8,5))
avg = df.groupby("City")["Rainfall (mm)"].mean()
avg.plot(kind="bar", color="orange")
plt.title("Average Rainfall by City")
plt.ylabel("Rainfall (mm)")
plt.show()
```

Output:







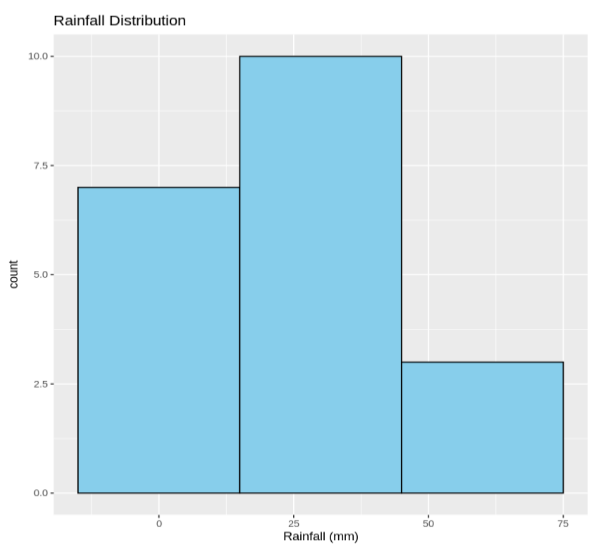


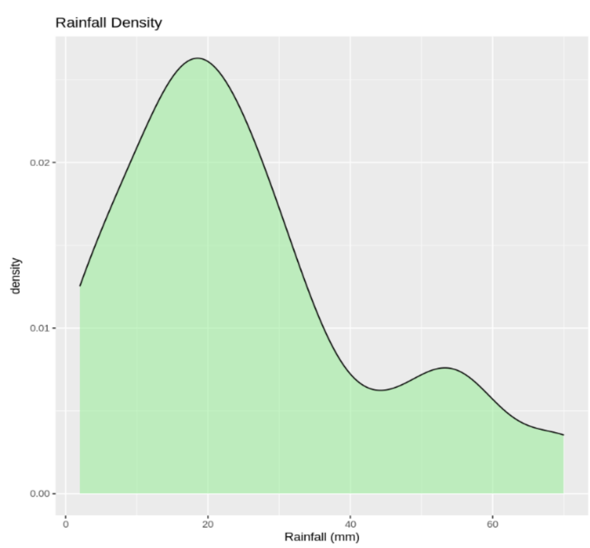
R code:

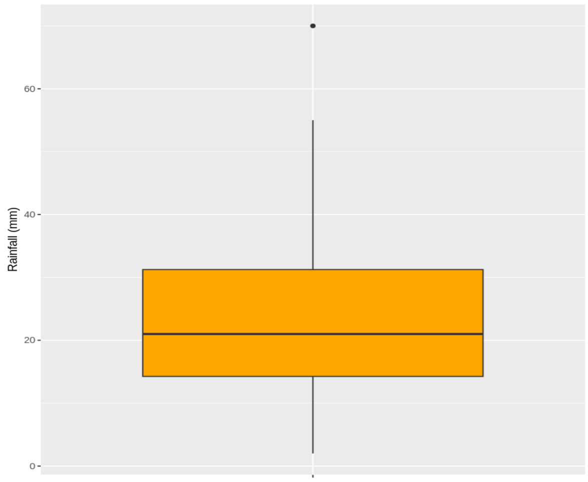
```
library(readxl)
library(ggplot2)
climate <- read_excel("Climate_Rainfall_Data.xlsx")
# 1. Histogram of Rainfall
```

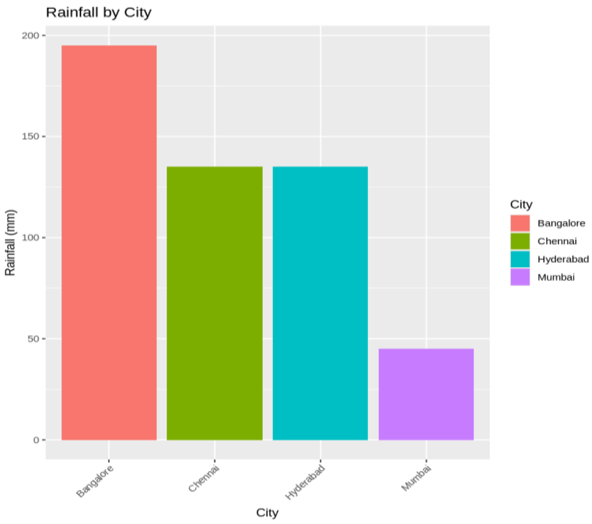
```
ggplot(climate, aes(x = `Rainfall (mm)`)) +
  geom_histogram(binwidth = 30, fill = "skyblue", color = "black") +
  ggtitle("Rainfall Distribution")
# 2. Density Plot of Rainfall
ggplot(climate, aes(x = `Rainfall (mm)`)) +
  geom_density(fill = "lightgreen", alpha = 0.5) +
  ggtitle("Rainfall Density")
# 3. Box Plot of Rainfall
ggplot(climate, aes(x = "", y = `Rainfall (mm)`)) +
  geom_boxplot(fill = "orange") +
  labs(x = "", y = "Rainfall (mm)")
# 4. Bar Chart of Rainfall by City
ggplot(climate, aes(x = City, y = `Rainfall (mm)`, fill = City)) +
  geom_bar(stat = "identity") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  ggtitle("Rainfall by City")
# 5. Histogram of Temperature
ggplot(climate, aes(x = `Temperature (°C)`)) +
  geom_histogram(binwidth = 2, fill = "red", color = "black") +
  ggtitle("Temperature Distribution")
```

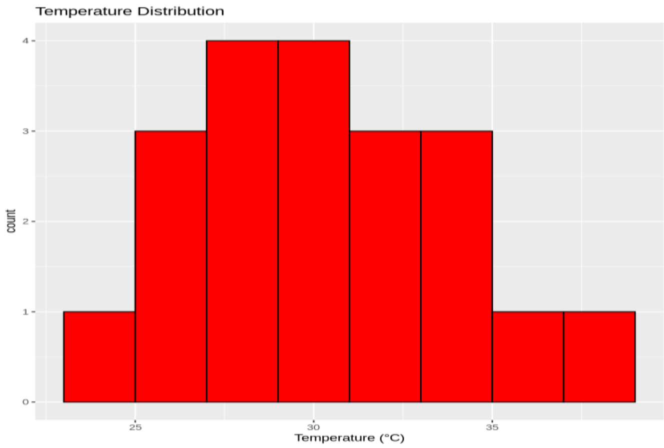
Output:











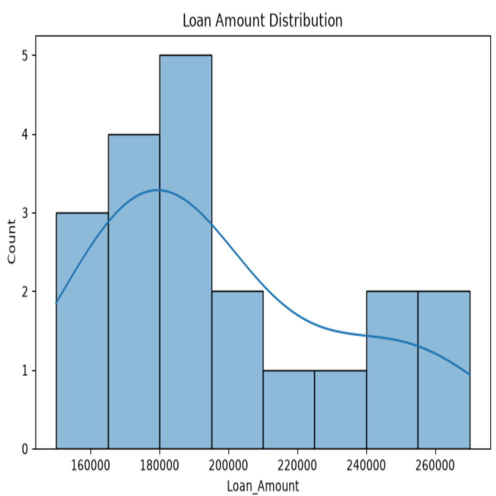
5. Bank Loan Risk Assessment

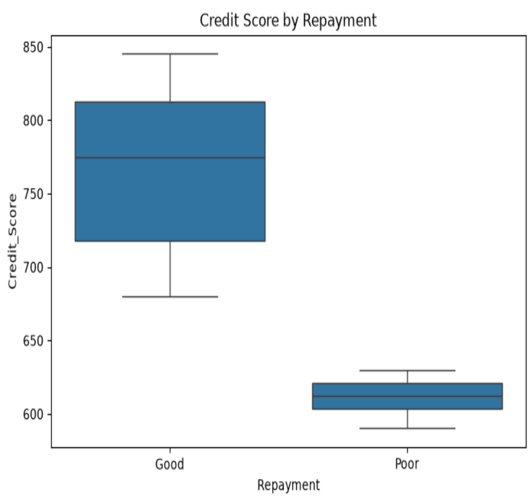
Python Code:

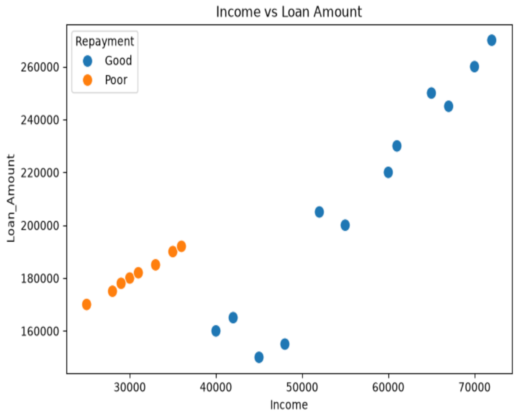
```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Read Dataset

df = pd.read_excel("Bank_Loan_Risk.xlsx")

# Histogram

plt.figure(figsize=(8,5))

sns.histplot(df["Loan_Amount"], bins=8, kde=True)

plt.title("Loan Amount Distribution")

plt.show()

# Box Plot

plt.figure(figsize=(8,5))

sns.boxplot(x="Repayment", y="Credit_Score", data=df)
```

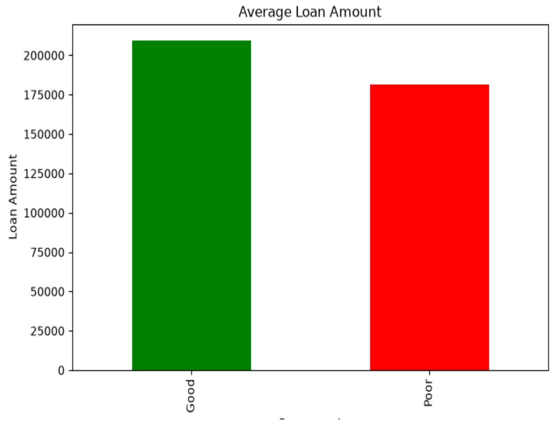
```
plt.title("Credit Score by Repayment")
plt.show()
# Scatter Plot
plt.figure(figsize=(8,5))
sns.scatterplot(x="Income", y="Loan_Amount",
            hue="Repayment", s=100, data=df)
plt.title("Income vs Loan Amount")
plt.show()
# Bar Chart
plt.figure(figsize=(8,5))
avg = df.groupby("Repayment")["Loan_Amount"].mean()
avg.plot(kind="bar", color=["green","red"])
plt.title("Average Loan Amount")
plt.ylabel("Loan Amount")
plt.show()
# Violin Plot
plt.figure(figsize=(8,5))
sns.violinplot(x="Repayment", y="Credit_Score", data=df)
plt.title("Credit Score Distribution")
plt.show()
```

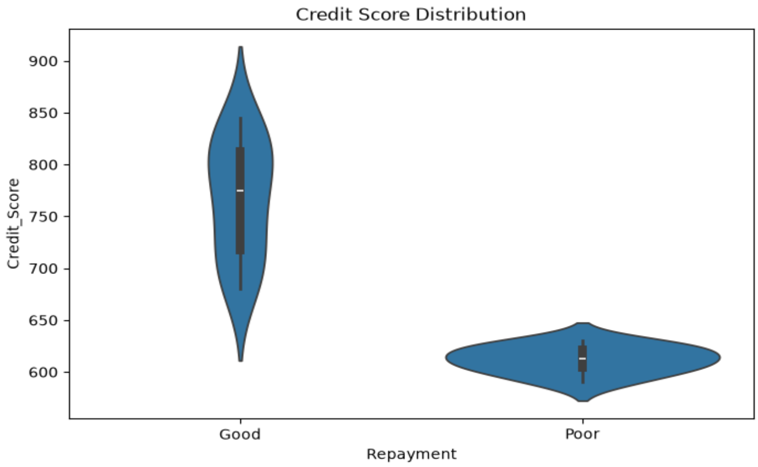
Output:









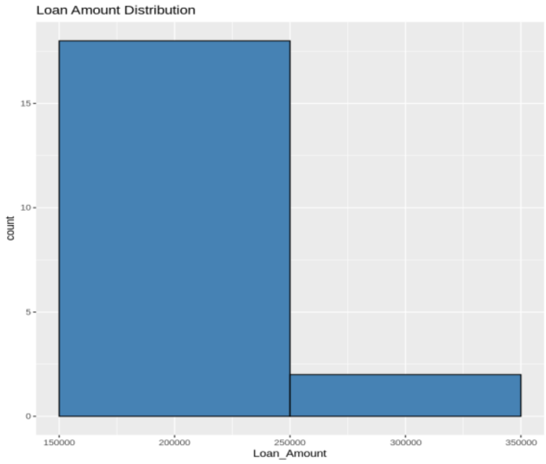


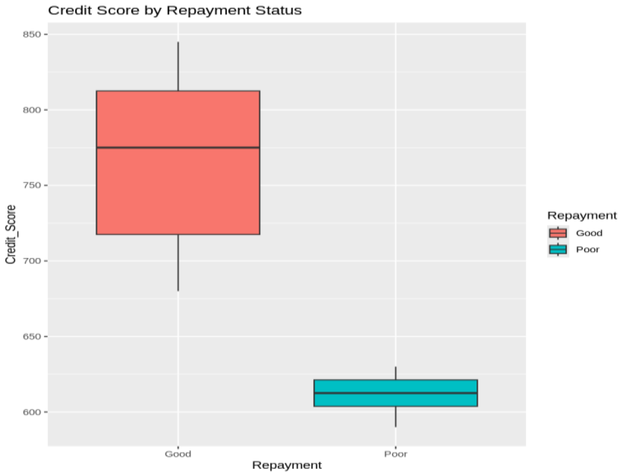
R code:

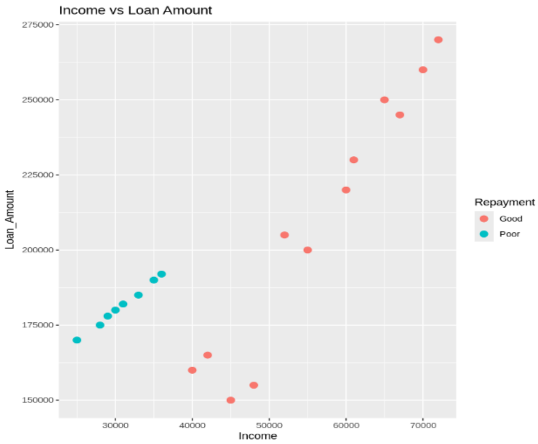
```
library(readxl)

library(ggplot2)

# Read Excel file

loan <- read_excel("Bank_Loan_Risk.xlsx")

# Check column names

print(colnames(loan))

# 1. Histogram - Loan Amount

ggplot(loan, aes(x = Loan_Amount)) +

  geom_histogram(binwidth = 100000,

                 fill = "steelblue",

                 color = "black") +

  ggtitle("Loan Amount Distribution")

# 2. Scatter Plot - Income vs Loan Amount

ggplot(loan,
```

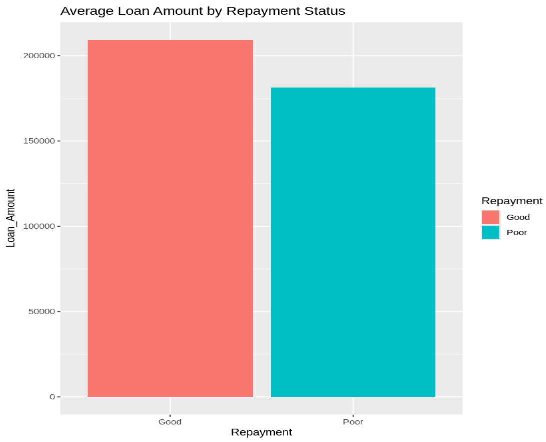
```r
      aes(x = Income,
          y = Loan_Amount,
          color = Repayment)) +
  geom_point(size = 3) +
  ggtitle("Income vs Loan Amount")
# 3. Box Plot - Credit Score
ggplot(loan,
      aes(x = Repayment,
          y = Credit_Score,
          fill = Repayment)) +
  geom_boxplot() +
  ggtitle("Credit Score by Repayment Status")
# 4. Bar Chart - Average Loan Amount
ggplot(loan,
      aes(x = Repayment,
          y = Loan_Amount,
          fill = Repayment)) +
  stat_summary(fun = mean,
              geom = "bar") +
  ggtitle("Average Loan Amount by Repayment Status")
# 5. Density Plot - Credit Score
ggplot(loan,
      aes(x = Credit_Score,
          fill = Repayment)) +
  geom_density(alpha = 0.5) +
  ggtitle("Credit Score Distribution")
```

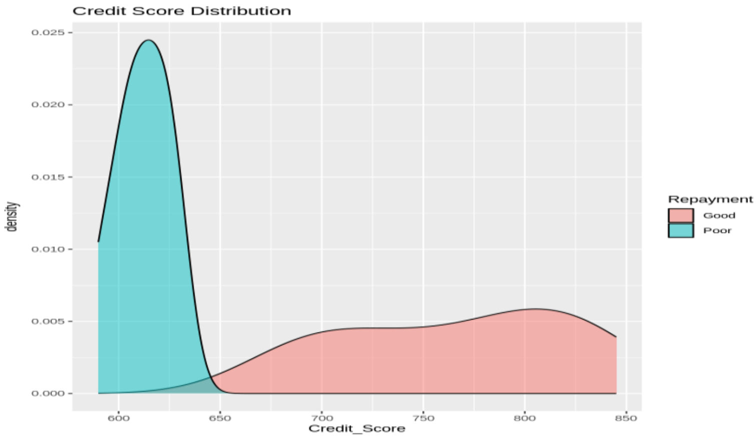
Output:


Loan Amount Distribution
count
Loan_Amount
150000
200000
250000
300000
350000
0
5
10
15


Credit Score by Repayment Status
Credit_Score
Repayment
Good
Poor
600
650
700
750
800
850


Income vs Loan Amount
Loan_Amount
Income
Repayment
Good
Poor
150000
175000
200000
225000
250000
275000
30000
40000
50000
60000
70000


Average Loan Amount by Repayment Status
Loan_Amount
Repayment
Good
Poor
0
50000
100000
150000
200000


Credit Score Distribution
density
Credit_Score
Repayment
Good
Poor
0.000
0.005
0.010
0.015
0.020
0.025
600
650
700
750
800
850